

Driveline Angle Adjustment



NOTICE: Suspension fasteners are critical parts because they affect performance of vital components and systems and their failure may result in major service expense. New parts must be installed with the same part numbers or equivalent part, if replacement is necessary. Do not use a replacement part of lesser quality or substitute design. Torque values must be used as specified during reassembly to make sure of correct retention of these parts.

NOTE: Some vehicles may exhibit a drive-away shudder or vibration under moderate to heavy acceleration from a stop, especially when heavily loaded or when towing a trailer. It is important to confirm how the vehicle is driven the majority of the time (loaded or unloaded), as adjusting the driveline angle in one condition (loaded or unloaded) may result in a shudder or vibration with the vehicle in the opposite condition (loaded or unloaded).

NOTE: Verify suspension is not modified from the original configuration. Aftermarket leveling kits and rear leaf spring shackle kits change ride height, which affects pinion angle and other critical driveline related operating angles. Vehicle must be at OEM specifications before accurate evaluation and diagnosis can be performed.

NOTE: The driveline angle must be measured with the vehicle in the same condition as the concern (loaded or unloaded) to determine the amount of adjustment necessary to correct the condition.

1. Measure the driveline angle. Refer to Driveline Angle Measurement.
- The U-joint operating angles must be equal or within one degree of each other.
 - The driveline can have a maximum 3.5 degree operating angle.
 - The U-joint operating angles must have at least one-half of one degree continuous operating angle.
 - If the driveline does not have an optimum 3.5 degree maximum operating angle, proceed to the following adjustment steps using the parts listed in the table below.

Part Number	Part Name
5A313	1 degree pinion angle shim
5A313	2 degree pinion angle shim
5595	Upper u-bolt center plate bolt (4 required)
5705	U-bolt (4 required)
W520215	U-bolt nut (8 required)
W713078	Center bearing support bolt
W713649	Center bearing support nut
4A209	Center bearing support shim
506545	Shock absorber lower bolt
W520214	Shock absorber lower nut

NOTE: The driveline angle on these vehicles can be changed by adding 1 degree or 2 degree pinion angle shims.

2. Remove and discard the shock absorber lower nut and bolt.

3. **NOTE:** To change the pinion angle.

Using a suitable jack to support the rear axle, remove and discard the 8 u-bolt nuts and the 4 u-bolts. Refer to Section 204-02.

4. **NOTE:** *Replace one bolt at a time.*

Replace the upper u-bolt center plate bolts. Refer to Section 204-02.

- Install the new bolts and tighten to 102 Nm (75 lb-ft).

5. **NOTE:** *Never use more than 1 shim on a side at a time.*

Install desired shim on the axle spring seat, or on top of the 4x4 spacer block.

- Lower the axle pinion using the thick edge of the shim forward, to decrease pinion angle.
- Raise the axle pinion using the thick edge of the shim rearward, to increase pinion angle.

6. Install 4 new u-bolts and 8 new u-bolt nuts. With the suspension at curb height, tighten the new u-bolt nuts evenly in a cross-type pattern in 4 stages. For additional information, refer to Section 204-02.

- Stage 1: Tighten in a cross pattern to 35 Nm (26 lb-ft).
- Stage 2: Tighten in a cross pattern to 70 Nm (52 lb-ft).
- Stage 3: Tighten in a cross pattern to 100 Nm (74 lb-ft).
- Stage 4: Tighten in a cross pattern to 133 Nm (98 lb-ft).

7. Install a new shock absorber lower nut and bolt. Refer to Section 204-02.

- Tighten the new nut and bolt to 90 Nm (66 lb-ft).

8. **NOTE:** *If adjusting axle pinion angle to reduce/eliminate driveline vibration or moan causes an issue with shudder on acceleration, then the pinion angle must be adjusted to optimize the balance between perceived vibration/moan and perceived shudder.*

NOTE: *Shims can be used to lower the center bearing support and minimize this condition. Up to 3 shims may be installed. The center bearing support bolts must be replaced using the part listed in the table or new grade 10.9 bolts. The bolts should be long enough to extend at least 3 threads past the nuts.*

Measure the driveline angle again. Refer to Driveline Angle Measurement.

- If necessary, remove and discard the 2 center bearing support bolts and install shims in 1/4 inch increments to lower the center bearing support.
- Install 2 new bolts and tighten to 48 Nm (35 lb-ft).